

Lab Assignment: Heuristic Search – Hill Climbing

Objective

The goal of this lab is to implement the Hill Climbing search algorithm to maximize a mathematical objective function. You will explore how local search strategies operate, visualize the search path, and analyze the impact of "local optima" on the success of the algorithm.

Background

Hill Climbing is an iterative local search algorithm that starts with a random solution and moves toward a neighboring solution with a higher value. The process repeats until no neighbor has a higher value than the current state.

Mathematically, given an objective function $f(x)$, we aim to find:

$$x^* = \arg \max_{(x)} f(x)$$

Problem Statement

Attempt to find the maximum value of a "rugged" function that contains multiple peaks.

- Objective Function: $f(x) = \sin(x^2) + \cos(x)$
- Search Space: $x \in [0, 5]$
- Step Size (Δx): 0.01

Task 1: Implementation

Implement the algorithm in Python. Your script should:

1. Define the objective_function(x).
2. Create a loop that checks the neighbors ($x + \Delta x$ and $x - \Delta x$).
3. Move to the neighbor only if it increases the value of $f(x)$.
4. Terminate when neither neighbor is better than the current state.

Task 2: Visualization

Generate a plot using a library like matplotlib.

1. Plot the function $f(x)$ as a smooth line.
2. Mark the Starting Point with a green dot.
3. Mark the Final Point (the peak found) with a red "X".
4. Trace the path taken by the algorithm.

Task 3: Comparison of Starting Points

Run the simulation twice with different initial values:

1. Trial A: $x_{\text{start}} = 0.5$
2. Trial B: $x_{\text{start}} = 4.0$

Compare the results. Did the algorithm find the same maximum in both cases?